

COMETS AND ASTEROIDS

COMETS AND ASTEROIDS ARE LEFTOVERS from when the nine planets formed in the Solar System 4.6 billion years ago. Comets are fragile balls of snow and dust found at the edge of the Solar System in the Oort Cloud. Some leave the cloud and travel towards the Sun. The Sun's heat melts the snow and the comet appears to grow in size many times over. Asteroids are made of rock and are found mainly between the orbits of Mars and Jupiter.

Anatomy of a comet

At the centre of a comet is the nucleus – a dirty ball of snow and dust that is just a few kilometres across. If a comet is close to the Sun, the snow becomes gas, and gas and dust are released, forming a vast cloud of material – the coma – and one or two tails.



Nucleus

Halley's Comet

Halley's Comet is the only comet that has been seen up close. Five spacecraft went to investigate when it travelled through the inner Solar System in 1986. The space probe *Giotto* took this image of its dark, potato-shaped nucleus.

Comet West passed Earth in 1976. It had two distinctive tails.

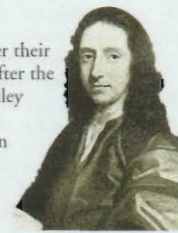


Thin, straight gas tail

Broad dust tail

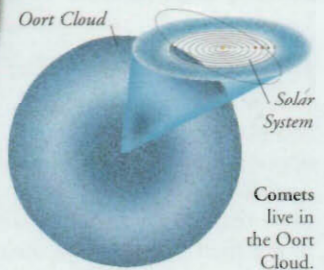
Edmond Halley

Comets are usually named after their discoverer, but one is named after the English scientist, Edmond Halley (1656–1742). He was the first person to show that comets can be periodic and follow orbits that return them again and again to Earth's sky.



Periodic comets

When a comet leaves the Oort Cloud it can travel on an orbit which returns it again and again to the inner Solar System. This is a periodic comet. About 150 are short-period comets; they return to appear in Earth's sky in periods of less than 200 years. Halley's Comet passes Earth every 76 years.



Oort Cloud

Surrounding the Solar System is the Oort Cloud, made up of 10 trillion comets. Although the cloud is large, it is so distant that the comets cannot be seen. Comets only become visible when they travel within the inner Solar System. Astronomers have seen about 700 comets in Earth's sky.

Meteoroids

Tiny pieces of dust and chunks of rock travel through space. They are meteoroids and originate from two main sources: comets and asteroids. About 220,000 tonnes of such material enter Earth's atmosphere a year. The smallest meteoroids produce meteors. Larger pieces reach Earth and land on its surface. These are meteorites.

Meteors

Tiny meteoroids burn up as they travel through Earth's atmosphere, producing streaks of light known as meteors. When Earth travels through a concentration of meteoroids, a meteor shower is produced. The meteoroid material is left by comets as they pass close to the Sun.

Meteorites

More than 3,000 meteorites land on Earth every year. Most fall in the sea, but a handful are seen to fall on land. There are three main types of meteorites: stony, iron, or stony-iron.



Fragment of nickel-iron

Stony-iron meteorite



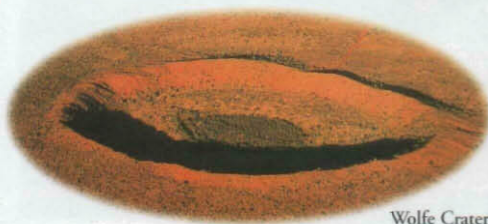
Nakhla meteorite

Rock from Mars

Eight meteorites are known to have come to Earth from Mars. The Nakhla meteorite fell in Egypt in 1911. It is 13 million years old. Such meteorites tell us that Mars once had running water.

Impact crater

When a meteorite lands on Earth, it can create a crater. Earth was once bombarded by meteorites but its surface has since changed, removing the evidence. Today, about 150 impact craters can be identified, including Wolfe Crater in Australia.



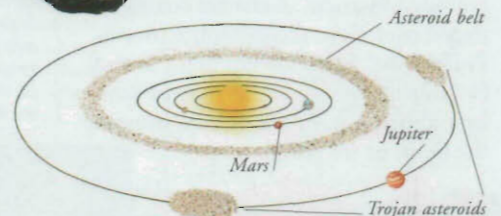
Wolfe Crater

Asteroid belt

Between the orbits of Mars and Jupiter is the asteroid belt – a doughnut-shaped ring made of millions of asteroids, pieces of rock, and metal. The smallest are specks of dust; the biggest, Ceres, is more than 900 km (550 miles) across.

Gaspra

About 5,000 asteroids have been identified, but only about 10 are spherical and larger than 250 km (150 miles) across. The smaller ones, such as the stony asteroid Gaspra, are irregular in shape.



Asteroid groups

Not all asteroids are in the asteroid belt. About 10 per cent travel in groups away from the belt. The Trojans travel along Jupiter's orbit, one group in front and one group behind the planet. The Amor, Apollo, and Aten groups all follow orbits closer to Earth.

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